

School of Computer Science and Information Technology

Department of Computer Science and Information Technology

Semester:	III
Specialisation:	SCT
Subject:	Programming for Cloud Platform

Activity - 1

Course with Project

Date of Submission:	27-09-2026
Submitted by:	Name: Devyansh Gupta USN No: 25MCAR0193
Faculty In-Charge	Dr. Pushpa J
Course:	HashiCorp Certified: Terraform Associate (003) Cert Prep by KodeKloud

Signature: _____

CERTIFICATE

This is to certify that Mr. Devyansh Gupta has satisfactorily completed the activity prescribed by JAIN (Deemed to be University) for the Third semester degree course in the year 2026. The activity is based on HashiCorp Certified: Terraform Associate (003) Cert Prep by KodeKloud and its accompanying Terraform cloud platform project.

Activity-1 Rubrics for MOOC Course Evaluation (Total: 15 Marks)

Sl.No	CRITERIA	MARKS	MARK Obtained
1	Course Completion on time	5	
2	Project Presentation	5	
3	Report Writing	5	
4	Total	15	

Date of Submission: 27-09-26

Signature of the Student Signature of the Faculty

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Student: Devyansh Gupta | USN: 25MCAR0193

1. COURSE SUMMARY WITH SCREENSHOT

HashiCorp Certified: Terraform Associate (003) Cert Prep by KodeKloud

This LinkedIn Learning course is an intermediate preparation course for the HashiCorp Certified: Terraform Associate (003) examination. It develops practical skills in infrastructure as code, Terraform CLI usage, providers, resources, variables, state, modules, import, workspaces, logging, and Terraform Cloud.

Course duration: 3 hours 10 minutes | Level: Intermediate | Completed by: Devyansh Gupta | Completion date: 20 September 2026



Contents

✓ Terraform taint
1m 56s video

✓ Logging
1m 42s video

✓ Terraform import
3m 3s video

✓ Terraform workspace
5m 16s video

7. Terraform Modules

✓ Terraform modules
5m 54s video

✓ Creating a module
8m 1s video

8. Terraform Cloud

✓ Terraform Cloud introduction
3m 7s video

✓ Terraform Cloud plans
3m 7s video

✓ Terraform Cloud demo
29m 18s video

Instructors



KodeKloud and Vijin Palazhi

2 instructors

Show all instructors

Related to this course

Certificates

Show all

Course details

3h 10m · Intermediate · Released: 6/14/2024

4.6 ★★★★★ (88)

In this course, learn the fundamentals and essential skills needed to fully prepare for the Terraform Associate (003) certification exam from HashiCorp. Instructor Vijin Palazhi helps you grasp the core concepts and practical techniques essential for mastering Terraform, a popular infrastructure as code (IaC) tool. Develop proficiency in creating and managing infrastructure resources efficiently using Terraform, and gain insights into infrastructure provisioning, configuration, and orchestration. Learn about IaC concepts, Terraform provider basics, variables and their usage in Terraform, managing Terraform state, using the Terraform CLI, and more.

Note: This course was created KodeKloud. We are pleased to host this training in our library.

Skills covered

DevOps

Terraform

Learners

Figure 1: Course content view showing Terraform modules and Terraform Cloud topics.

2. COURSE LEARNING OUTCOMES

Infrastructure as Code

Understand why infrastructure can be represented as version-controlled, repeatable configuration.

Terraform workflow

Use init, fmt, validate, plan, apply, show, output, and destroy as part of a controlled workflow.

State management

Explain Terraform state, state locking, remote backends, drift, and the role of Terraform Cloud.

Reusable configuration

Use variables, outputs, modules, expressions, and provider configuration to build maintainable projects.

Operational confidence

Read a plan, identify resource dependencies, import existing resources, and troubleshoot with logging.

The final project applies these outcomes by defining a secure, repeatable AWS static website platform with S3 and CloudFront resources, outputs, provider constraints, and automated validation.

3. COURSE COMPLETION CERTIFICATE

Course: LinkedIn Learning - HashiCorp Certified: Terraform Associate (003) Cert Prep by KodeKloud

Learner: Devyansh Gupta

Completed: Sep 20, 2026 at 06:50 PM UTC

Duration: 3 hours 10 minutes

Top skills covered: DevOps, Terraform

Certificate ID: 98f20bbe44e7daad5af0259b59fc24ebc3e37489225625644fb6854138dd6646

The source certificate PDF supplied with this report is the evidence for course completion and is retained separately from this project report.

4. PROJECT SUMMARY - PROBLEM STATEMENT

Problem statement

Manual creation of a website hosting platform is repetitive, difficult to audit, and prone to configuration drift. The project solves this by defining the infrastructure as Terraform code that can be reviewed, validated, versioned, and reproduced.

Project objective

Provision a secure static website platform on AWS using a private S3 bucket as the origin and an Amazon CloudFront distribution as the HTTPS delivery layer. The design prevents direct public access to the bucket and grants read access only to CloudFront through Origin Access Control.

Scope

The repository contains provider and version constraints, variables, outputs, S3 resources, CloudFront resources, an IAM bucket policy, a sample website, GitHub Actions validation, documentation, and a reproducible local validation record.

4.1 PROJECT ARCHITECTURE

Layer	Component	Purpose
Source	GitHub repository	Version control, collaboration, and proof link
IaC	Terraform	Declarative resource definitions and dependency graph
Storage	Amazon S3	Private bucket for website assets
Delivery	Amazon CloudFront	HTTPS distribution and caching
Security	Origin Access Control + bucket policy	Only CloudFront can read the S3 origin
Automation	GitHub Actions	Format, initialize, and validate on pull requests and pushes

Flow: Developer -> GitHub -> Terraform plan -> AWS S3 + CloudFront -> HTTPS website

4.2 STEPS OF IMPLEMENTATION

1. Create Terraform version and AWS provider constraints.
2. Define configurable region, project name, environment, and CloudFront price class variables.
3. Create an S3 bucket with public access blocked and versioning enabled.
4. Create a CloudFront Origin Access Control and distribution with HTTPS redirection.
5. Create an IAM policy document and attach it to the S3 bucket so CloudFront can read objects.
6. Expose bucket, distribution ID, and CloudFront domain through Terraform outputs.
7. Add a sample website and GitHub Actions workflow for formatting and validation.
8. Run `terraform fmt -recursive`, `terraform init -backend=false`, and `terraform validate`.
9. Commit and upload the project to GitHub for repeatable review and submission proof.

Validation result: Terraform v1.15.9 reported that the configuration is valid. Applying infrastructure was intentionally not performed because it requires AWS credentials and creates billable cloud resources.

4.3 OUTPUT

Expected Terraform outputs after a successful AWS apply:

website_bucket_name - The private S3 bucket containing website assets.

cloudfront_distribution_id - The CloudFront distribution identifier.

cloudfront_domain_name - The HTTPS domain name used to access the website.

Local evidence

The repository includes the site/index.html landing page and the GitHub Actions validation workflow. The Terraform configuration has been formatted and validated successfully. The deployment remains ready for an AWS-authenticated plan/apply workflow.

Security outcome

The S3 bucket blocks public access. CloudFront uses Origin Access Control with SigV4 signing, and the bucket policy limits object reads to the CloudFront service principal and distribution ARN.

5. PROOF LINK: GITHUB

Repository: <https://github.com/Devyansh-Gupta/terraform-cloud-platform-devyansh-25mcar0193>

Repository visibility: Private

Commit: Create Terraform cloud platform project

The repository contains the complete Terraform source, sample website, workflow, README, provider lock file, and report-generation source. The private setting protects the student identifier while allowing the owner to share access with the evaluator.

Suggested evaluator commands

```
terraform init
```

```
terraform fmt -check -recursive
```

```
terraform validate
```

```
terraform plan -var="project_name=devyansh-terraform-demo" -var="environment=dev"
```

6. CONCLUSION

The LinkedIn Learning course provided the concepts required to design and operate Terraform configurations. The project applies those concepts in an end-to-end cloud platform: infrastructure is declared in code, secured through least-privilege access, exposed through outputs, checked automatically, and published in Git for review.

The implementation is ready for a credentialed AWS plan and apply. Keeping the apply step behind an explicit approval protects the account from unexpected cost or resource creation while preserving a complete, reproducible submission.

Submitted by: Devyansh Gupta

USN: 25MCAR0193

Date of Submission: 27-09-26

7. REFERENCES

1. LinkedIn Learning. HashiCorp Certified: Terraform Associate (003) Cert Prep by KodeKloud. <https://www.linkedin.com/learning/hashicorp-certified-terraform-associate-003-cert-prep-by-kodekloud/>
2. HashiCorp Terraform documentation. <https://developer.hashicorp.com/terraform/docs>
3. HashiCorp AWS provider documentation. <https://registry.terraform.io/providers/hashicorp/aws/latest/docs>
4. Supplied course completion certificate: CertificateOfCompletion_HashiCorp Certified Terraform Associate 003 Cert Prep by KodeKloud.pdf
5. Supplied reference report: PCP_25MCAR0144_DHRUVI.pdf